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B. E. - (Mechanical) (Sem-VII) (Revised Course 2019-2020)

EXAMINATION MAY 2024

ME 725 Additive Manufacturing (OPEN ELECTIVE)

[Time: 3:00 Hours]

[Max. Marks: 100]

- Instructions:** 1) Answer **any two** questions each from **Part A** and **Part B** and **one** question from **part C**
- 2) Assume missing data, if any
- 3) Illustrate your answers with neat sketches wherever required.

PART A

Answer any **TWO** questions from the following:

- Q1 (A) Differentiate with an example additive and subtractive manufacturing process. (06)
- (B) Explain the role of Additive manufacturing in new product development with the aid of an example (07)
- (C) Suggest the additive manufacturing process for the crystal Manufacturing unit to take trials of transparent artificial diamond made up from liquid polymer. Justify your answer with the working principle, advantages & disadvantages. (07)
- Q2 (A) Explain with an aid of example the difference between additive manufacturing, rapid tooling and 3D printing. (06)
- (B) A shipyard is looking for an additive manufacturing process which can print five dimensional propellers from titanium material. Suggest the process with the aid of a neat sketch, working principle, advantages & disadvantages. (07)
- (C) Suggest additive manufacturing process to build prototypes using paper as a raw material with the aid of a neat sketch, working principle, advantages & disadvantages. (07)

- Q3 (A) Define additive manufacturing process and explain in detail the steps followed in the additive manufacturing from concept to prototype. (06)
- (B) Suggest the Cost-effective additive manufacturing process for the ornamental gift manufacturing Company to develop products from thermoplastic materials. Justify your answer with the working principle, advantages & disadvantages. (07)
- (C) A Medical firm would like to manufacture orthopaedic Implants from metal powder with an accuracy of 1 Micron. Suggest the process with the aid of a neat sketch, working principle, advantages & disadvantages. (07)

PART B

Answer any **TWO** questions from the following:

- Q4 (A) Explain the concept of spray metal tooling with a neat sketch (04)
- (B) Explain with a neat sketch the process of silicon rubber tooling. (08)
- (C) Write a note on the factor which influences the accuracy of the additive manufactured component. (04)
- (D) Differentiate between soft & hard tooling (04)
- Q5 (A) Write a note on magic and mimic software used for the Rapid Prototyping (04)
- (B) Explain with a neat sketch the process of Aluminium filled epoxy tooling (08)
- (C) Write a note on (08)
- Part building errors and Error in finishing
 - factors influencing accuracy and Slicing
- Q6 (A) Explain the Process of Reverse engineering with a suitable example. (06)
- (B) Explain with a neat sketch the quick cast process (07)
- (C) Explain the concept of part orientation and support generation for the following 3D part (07)



PART C

Answer any **ONE** question from the following

- Q7 (A) Explain the different process parameters which govern the Mechanical strength of Prototypes made from Selective laser sintering process (06)
- (B) Explain with a neat sketch the process of sand casting tooling (07)
- (C) Explain the process of slicing the STL file and the list the software's used for slicing, (07)
- Q8 (A) Explain with a neat sketch the working of Solid Ground Curing stating its advantages & disadvantages over other processes (07)
- (B) Explain with a neat sketch the working of Electron Beam Melting Stating its advantages & Disadvantages over other processes (07)
- (C) List and explain different types of materials used for additive manufacturing (06)